

STAKEDXF

UI & CAPABILITIES · v1.24

Rugged on-device field kit — Civil 3D DWG → Trimble DXF recovery and scaled staking-plot PDFs on the TSC5 handheld.

ENGINE

LIBREDWG · EZDXF · FLUTTER

OUTPUT

DXF R2010 · PDF

TARGET

TRIMBLE TSC5

AGENDA

- Before / after — visible break from v1.19
- Design system
- Home / operations
- CONVERT · DWG → DXF pipeline
- PLOT · Export points + staking sheet
- Plot customization
- Staking plot examples
- Install on Trimble TSC5
- Field workflow end-to-end

BEFORE · AFTER

v1.19 — retired

StakeDXF

≈ Convert DWG → DXF

Convert for Trimble Access

Soft cards. Rounded chrome. Forest wash. Marketing copy.
Every version prior to v1.22 looked essentially the same.

v1.24 — shipped

STAKEDXF

FIELD-KIT / TSC5 / TRIMBLE

01

CONVERT

DWG → DXF · Civil 3D linework recovery

→

RUN CONVERT

→

Hard 90° corners. Hairline rules. Mono telemetry. Safety-orange reserved for commands.
Corner brackets frame the active field of view. Reads as a piece of ruggedized field kit — not an app.

DESIGN SYSTEM

BG

#07090B

CARD

#10131A

RAIL

#1D212B

BORDER

#252B36

FG

#F1F4F7

DIM

#C6CAD1

MUTED

#798594

ACCENT

#FF5A1F

OK

#7EDF9F

WARN

#F1C15A

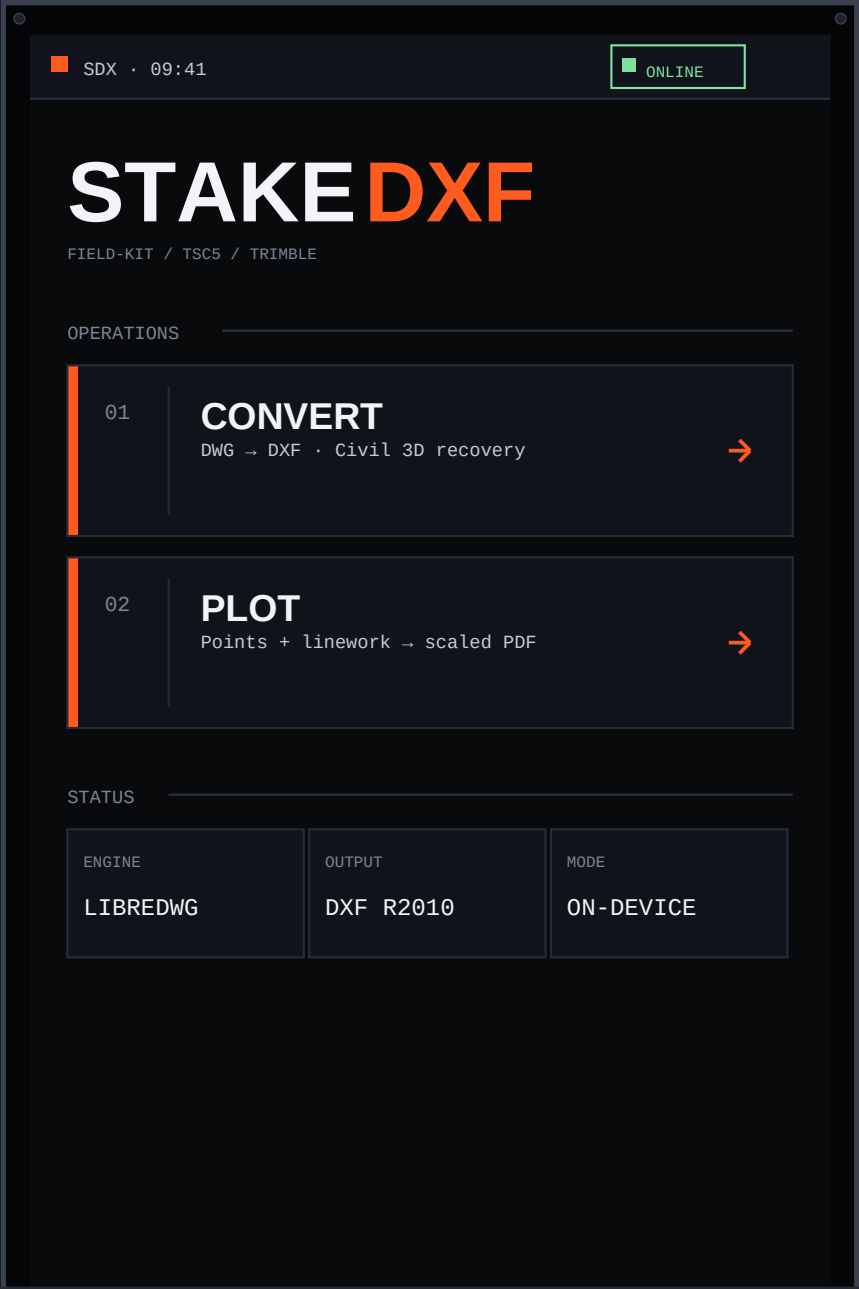
DANGER

#E4474D

PRINCIPLES

- Hard 90° corners — no rounded chrome
- Mono telemetry for coordinates, versions, entity counts
- Safety-orange only for active / primary actions
- Hairline rules over cards; content over decoration
- Minimum 44 px tap targets for TSC5 field use

HOME / OPERATIONS



Two ops, one screen: CONVERT and PLOT.

Rugged action rails with a numeric tag, thick left rail, and arrow — reads like a hardware toggle, not a marketing card.

TELEMETRY

- Ribbon: SDX build + ONLINE / LOCAL pills
- Status grid: ENGINE / OUTPUT / MODE readouts
- Footer: TRIO FIELD OPS · NO CLOUD · NO TRACKING

CONVERT · DWG → DXF

SDX · 09:41

ONLINE

CONVERT / DWG → DXF

DRAWING

ALPINE_HILLS.DWG

TRIO/PROJECTS/ALPINE/...

RUN CONVERT

→

CONVERT / OK

RECOVERED 390 ENTITIES · 12 LAYERS

STAKEABLE390

LAYERS12

PROXIES148

FILEALPINE_HILLS_TRIMBLE.DXF

LAYERS3/12ALL

✓

P-CURB390

✓

P-U-STM649

✓

P-CL124

DEFPOINTS26804

HATCH33

SAVE DXF · 3 LAYERS

→

- Input slot glows orange when a drawing is loaded
- RUN CONVERT primary button with arrow slot
- Result readout tagged CONVERT / OK with mono KVs
- Layer checklist with hard tick boxes + entity counts
- SAVE DXF button carries the selected-layer count

PLOT · EXPORT POINTS — START

SDX · 09:41

ONLINE

PLOT / EXPORT POINTS

JOB

ALPINE_HILLS

IMPORT POINTS CSV

LINK DXF LINEWORK

IMPORT A PNEZD CSV
FROM TRIMBLE ACCESS TO BEGIN.

- Enter a JOB name — becomes the sheet title
- IMPORT POINTS CSV / TXT from Trimble Access
- Optional LINK DXF LINEWORK for on-plot detail
- PNEZD headered or headerless supported

PLOT · CUSTOMIZE + CREATE

SDX · 09:41

ONLINE

PLOT / EXPORT POINTS

JOB

ALPINE_HILLS

IMPORT POINTS CSV

LINK DXF LINEWORK

PLOT OPTIONS

MARKER

LARGE X

LABEL

NUM + ELEV

SCALE

1"=40'

SHEET

ANSI B 11×17 LAND.

LAYERS 3/3

CL

CURB

STRUCT

- MARKER · triangle / X / cross / circle / dot
- LABEL · number / desc / elev combinations
- SCALE · engineering presets or custom 1"=N'
- SHEET · ANSI A–D, portrait or landscape
- LAYERS · tick only the linework you want
- Object library places hydrants / MH / signs
- CREATE STAKING PLOT PDF or EXPORT CSV

LAYER PROPERTIES MANAGER

SDX · 09:41

ONLINE

LAYER PROPERTIES MANAGER

5 / 12 ON

FILTER

ALL

ON

OFF

LOCKED

OVERRIDDEN

Filter layers...

NAME	ENT	On	Frz	Lk	COLOR	LTYPE	LW	TR	LTS
P-CURB	390					DASH	0.35	30	1.0
P-U-STM	649					CONT	0.50	20	1.0
P-CL	124					DASHD	0.25	0	1.5
P-BLDG	82					CONT	0.70	0	1.0
HATCH	33					HIDD	0.18	40	1.0
DEFPOINTS	26804					DOT	0.18	0	1.0
P-VP	4					CONT	0.35	0	1.0
P-ROAD	260					CONT	0.50	0	1.0

GLOBAL

LTS

1.0

TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

The DXF linework grid is a true Layer Properties Manager — Civil 3D column order, sortable headers, sticky Name pane, horizontally scrolling data strip.

COLUMNS

- On — light-bulb toggle (draws layer on the plot)
- Frz — freeze marker (snowflake / sun icon)
- Lk — padlock; locked layers can't be drag-selected
- Color — ACI / CTB / true-color HSV picker
- Linetype — CONT / DASHED / DASHDOT / HIDDEN ...
- LW — printed weight in points
- Tr — transparency %; opacity = 100 - Tr
- LTS — per-layer linetype scale

TOOLBAR + FILTER

- All-on / all-off / invert bulbs
- Lock-all / unlock-all
- Reset overrides for selected layer or every layer
- Chips: ALL · ON · OFF · LOCKED · OVERRIDDEN
- Live search box — filters by layer name

COLOR PICKER · ACI + CTB

- Full ACI 1 – 255 grid, CTB-resolved swatches
- Header shows the current color swatch
- Selected chip framed in safety-orange
- Tap ByLayer / CTB default to clear override
- Two-tab sheet — ACI on the left, True on the right

SDX · 09:41

ONLINE

LAYER COLOR — P-CL

ACI / CTB

True color

ACI NUMBER

34

SET

APPLY

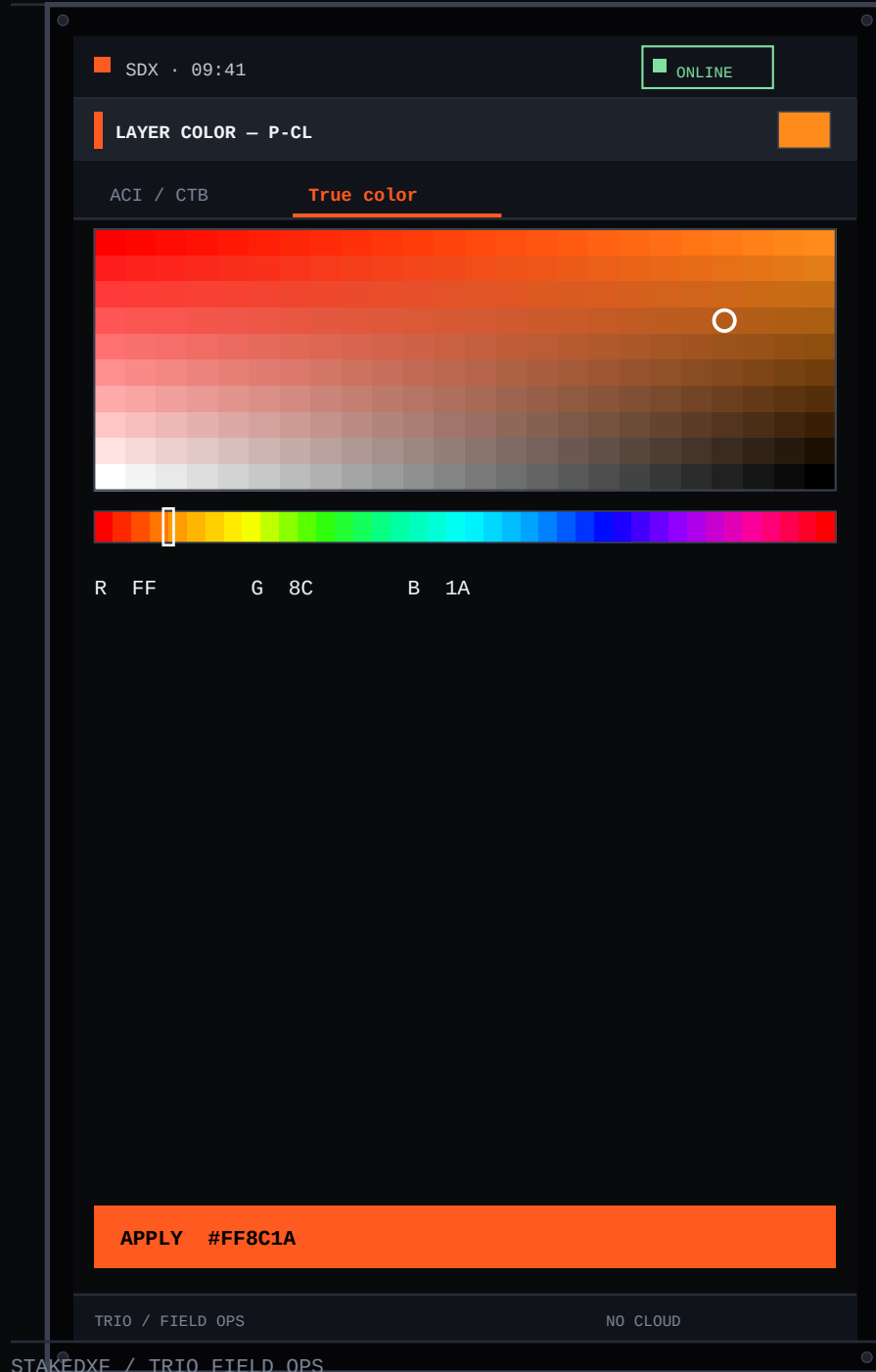
TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

COLOR PICKER · TRUE COLOR

- Sat/Value gradient — pick with a single tap
- Full-hue slider under the pad
- Live R / G / B readout in mono
- APPLY writes the exact #RRGGBB override



LINETYPE PICKER

SDX · 09:41

ONLINE

LINETYPE — P-CL

CONTINUOUS

DASHED

DASHED2

DASHDOT

DASHDOT2

HIDDEN

HIDDEN2

DOT

DOT2

BORDER

CENTER

PHANTOM

BYLAYER DEFAULT

TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

- Full LIN catalog + user-loaded linetypes
- Each row shows the actual dash pattern preview
- Tap ByLayer default to inherit the layer style
- Choice applies to the whole layer — same as Civil 3D

LINEWEIGHT PICKER

SDX · 09:41

ONLINE

LINEWEIGHT (pt) — P-CL

0.18

0.25

0.35

0.50

0.70

1.00

1.40

2.00

BYLAYER / CTB DEFAULT

TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

- Standard AutoCAD weights 0.18 – 2.00 pt
- Each row shows a scaled preview stroke
- Choice affects the printed PDF stroke only
- Tap ByLayer / CTB to defer to the CTB table

TRANSPARENCY / OPACITY

SDX · 09:41

ONLINE

TRANSPARENCY — P-CL

OPACITY 75% · TRANS 25%

RESET

APPLY

TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

- Live-updates the plot preview while dragging
- OPACITY and TRANS both shown in mono
- RESET returns to the resolved ByLayer / CTB value
- APPLY commits the override; no more no-op slider

SDX · 09:41

ONLINE

LINETYPE SCALE — P-CL

1.50

RESET

APPLY

TRIO / FIELD OPS

NO CLOUD

- Per-layer LTS override — 0.1 to 20.0
- Global LTS lives on the LPM footer
- Resolved value = layer LTS × global LTS
- Reset returns to the resolved 1.0

LINEWORK PROPERTIES

SDX · 09:41

ONLINE

LINEWORK · P-CURB · LWPOLYLINE

SEG 3 / 12 · 53.42 LF

LAYER	P-CURB
COLOR	ACI 1 RED
LINETYPE	DASHED
WEIGHT	0.35 pt
SCALE	1.0
TRANS	30 %
SEG LEN	53.42 ft

TRIM SEGMENT

DELETE SEGMENT

CLEAR SELECTION

TRIO / FIELD OPS

NO CLOUD

- Selecting a segment on the preview opens this panel
- Read-outs: layer / color / linetype / weight / scale
- TRIM SEGMENT — surgical edit without a full delete
- DELETE SEGMENT — drop just this run
- CLEAR SELECTION restores the sticky behaviour fix

POINT PROPERTIES

SDX · 09:41

ONLINE

POINT · #142

#142 EL 1218.42

PT #	142
NORTH	6,241,822.145
EAST	1,842,633.911
ELEV	1218.42 ft
DESC	CURB RETURN
LABEL	# + ELEV
MARKER	LARGE X
OFFSET	dN 0.20 dE 0.15

MOVE LABEL

HIDE POINT

RESET OFFSET

TRIO / FIELD OPS

NO CLOUD

- Full PNEZD read-out + label offset
- Marker + label style overridable per-point
- MOVE LABEL — drag on the preview to reposition
- HIDE POINT — omit from the final PDF
- RESET OFFSET returns to the auto-layout position

TEXT STYLE PICKER

SDX · 09:41

ONLINE

TEXT STYLE

STANDARD

h 3.0 mm · 0.90 W

142 ELEV 1218.42

ROMANS

h 2.4 mm · 0.90 W

142 ELEV 1218.42

ROMAND

h 2.5 mm · 0.90 W

142 ELEV 1218.42

ITALIC

h 2.4 mm · 0.90 W

142 ELEV 1218.42

MONO

h 2.2 mm · 0.90 W

142 ELEV 1218.42

SANS-B

h 3.2 mm · 0.90 W

142 ELEV 1218.42

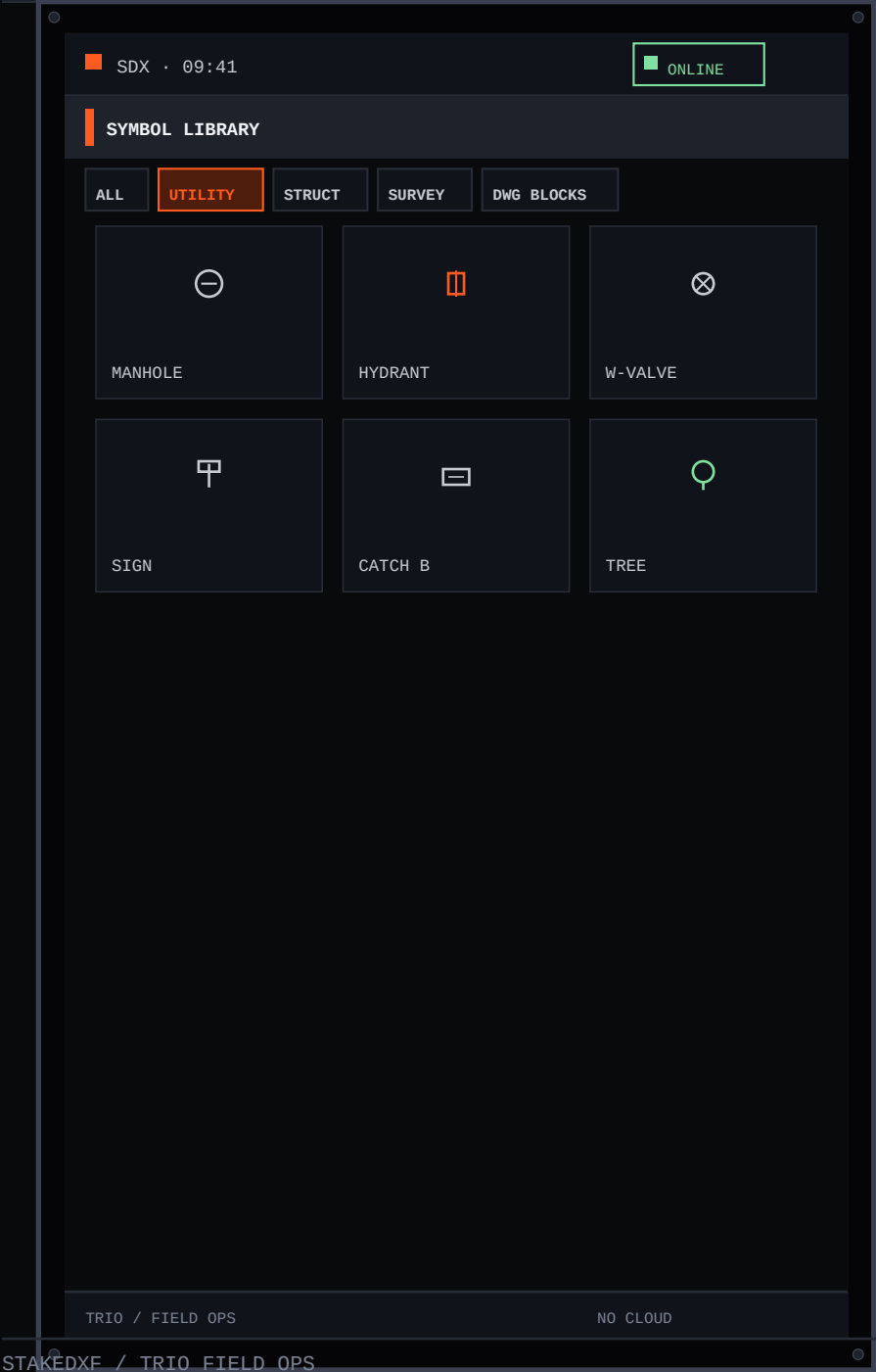
TRIO / FIELD OPS

NO CLOUD

STAKEDXE / TRIO FIELD OPS

- Every SHX-equivalent style shown with a live preview
- Height / width factor labelled in monospace
- STANDARD, ROMANS, ROMAND, ITALIC, MONO, SANS-B
- Applies to the selected label / callout style

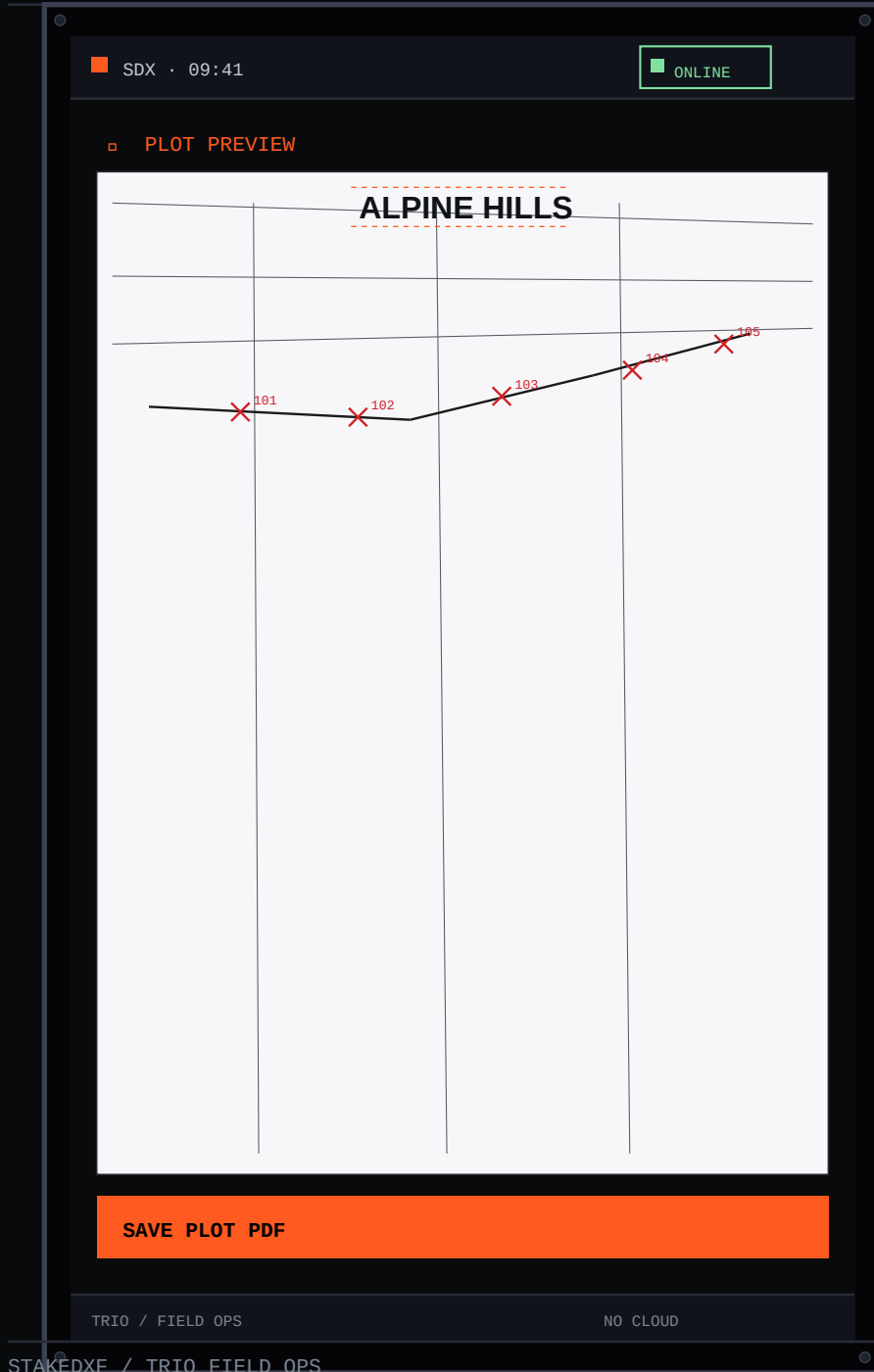
SYMBOL LIBRARY



- Category chips: ALL / UTILITY / STRUCT / SURVEY / DWG BLOCKS
- Built-ins + AutoCAD blocks pulled from the DWG catalog
- Tap a card, then place on the sheet at real-world scale
- Symbols honour marker size + layer color overrides

PLOT PREVIEW · ANSI FULL BLEED

- Full-bleed sheet — linework runs edge-to-edge
- No border, no cream panel, no title block band
- Plot title is OPTIONAL — off by default, draggable when on
- Red X stake markers with numbered leader labels
- Sub-menus cap at ~60% height so the preview stays visible
- SAVE PLOT PDF writes the final Trimble-ready sheet



SHEET TEMPLATE PICKER

- Single layout — ANSI FULL BLEED
- Pick ANSI A / B / C / D
- Portrait or landscape
- Active template highlighted in safety-orange
- Plot is pure full-bleed — nothing but the plan by default

PLOT CUSTOMIZATION

MARKERS

- Filled triangle · Triangle outline
- Cross (+) · X · Large X
- Circle · Dot · Large dot

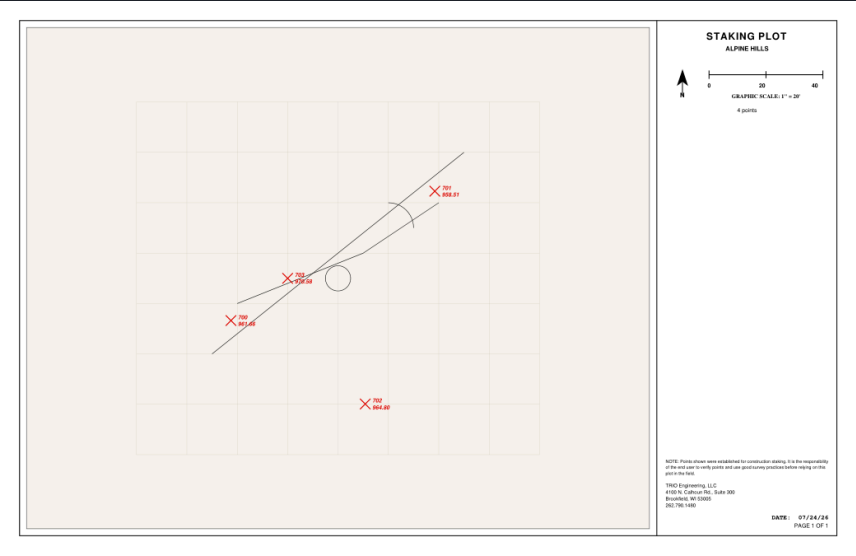
LABELS

- Point number
- Number + description
- Number + elevation
- Number + description + elevation
- No labels

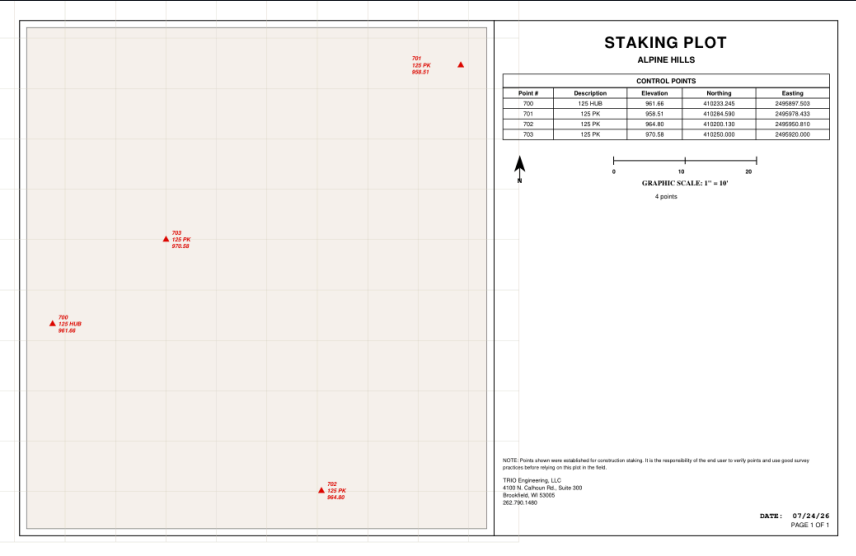
SHEET & LINEWORK

- Plot title: optional, draggable & resizable on preview
- No bounding box, no scale text, no north arrow, no callout
- No hatch fills on solid objects — clean outlines
- Linked DXF: LINE / LWPOLYLINE / ARC / CIRCLE by layer
- Auto engineering scale or hard-set 1"=N'
- Layer LOCK column prevents accidental drag-selection
- Full ACI + CTB + true-color HSV picker

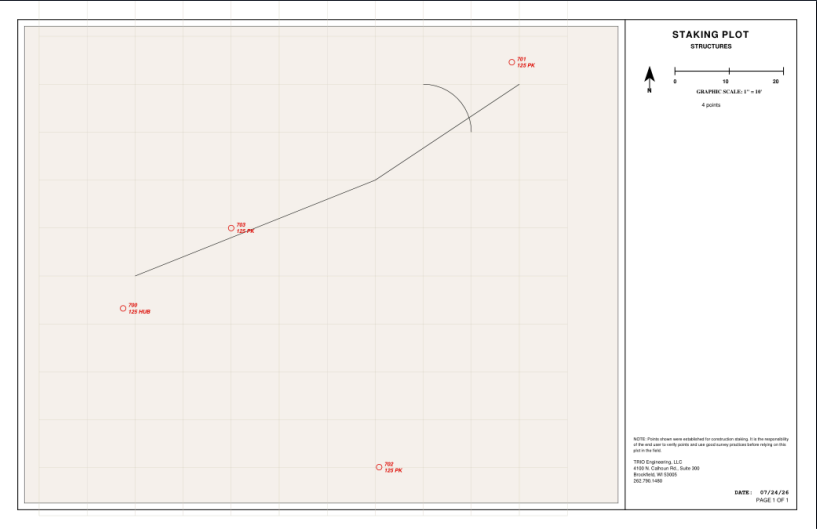
STAKING PLOT EXAMPLES



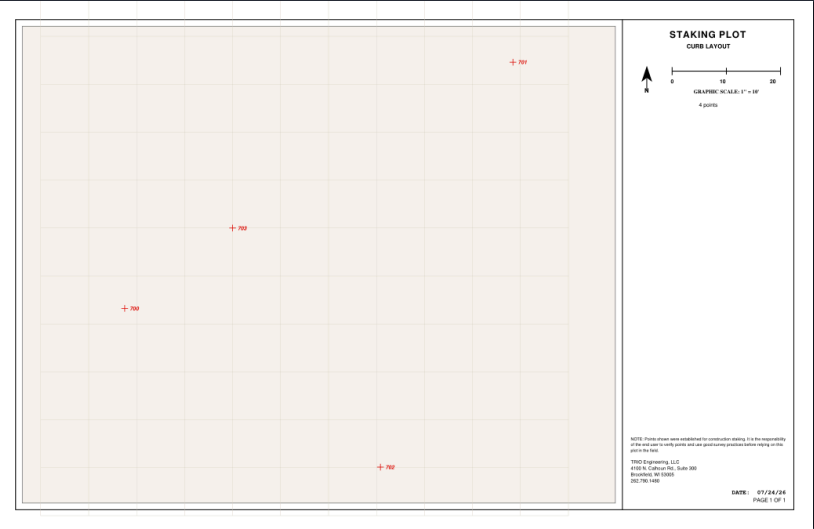
FIELD STAKING · LARGE X + ELEV



CONTROL-NOTE STYLE + TABLE

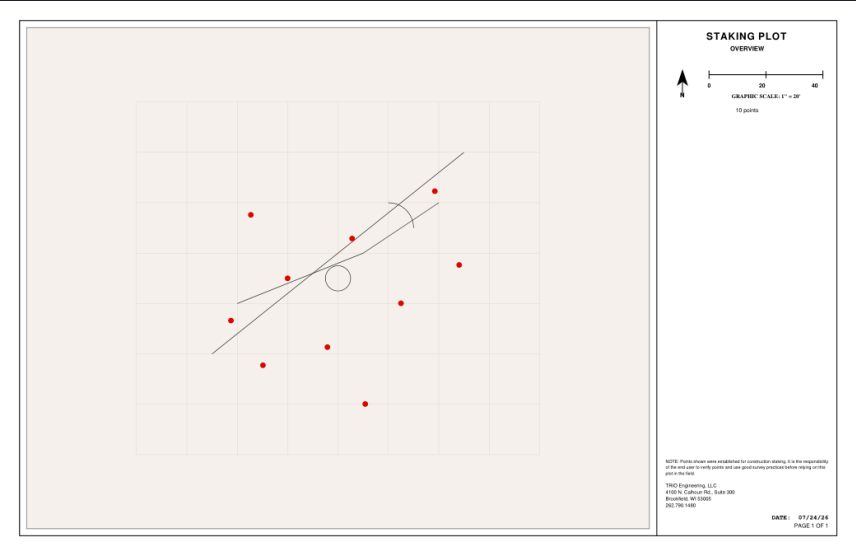


CIRCLES + CURB LAYER

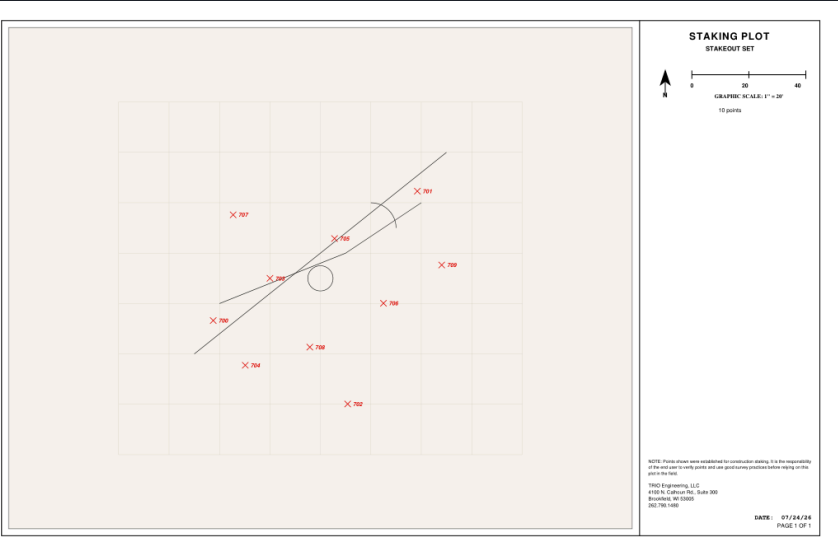


CROSS (+) NUMBER ONLY

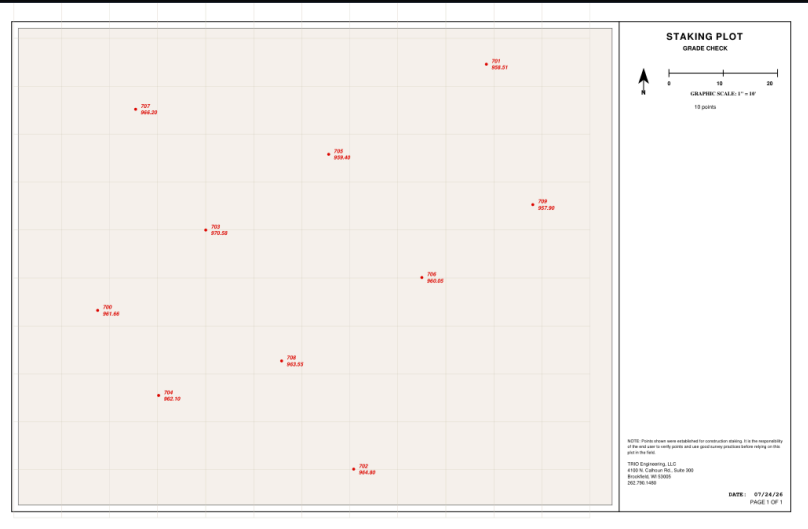
STAKING PLOT EXAMPLES



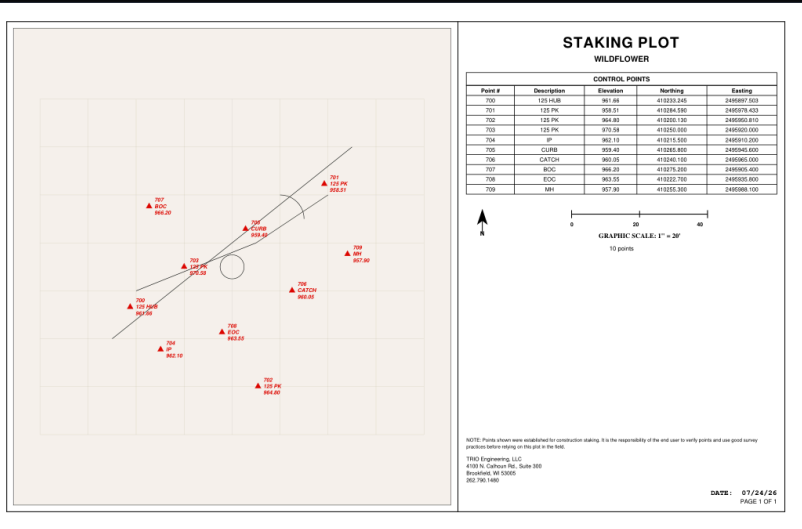
OVERVIEW · DOTS · NO LABELS



DENSE SET · NUMBER ONLY



GRADE CHECK · ELEV LABELS



FULL SHEET + LINEWORK

INSTALL / TRIMBLE TSC5

- Copy dist/StakeDXF vX.Y.Z.apk onto the TSC5
- Open the APK — allow Install from this source
- Open StakeDXF from the app drawer
- MDM blocked? Ask IT for unknown-source exception
- APK is Android-only — will not install on iPhone

FIELD WORKFLOW

- Office: save Civil 3D DWG with proxies preserved
- TSC5: CONVERT → save DXF into Trimble job folder
- Trimble Access: Map files → selectable → Stakeout
- Export points CSV from Access for shots you need
- StakeDXF PLOT → customize → CREATE staking plot PDF
- Keep PDF open while staking for a scaled field sheet

DOCS & HELP

- [dist/INSTALL_TSC5.md](#) — quick install card
- [dist/StakeDXF_User_Guide.pdf](#) — full usage / help
- [docs/USER_GUIDE.md](#) — same guide in Markdown
- [dist/plot_examples/](#) — sample staking PDFs
- [PR #18](#) — source, APK, release notes

READY FOR THE FIELD.

STAKEDXF · TSC5

RECOVER LINEWORK. PLOT POINTS. STAKE WITH CONFIDENCE.